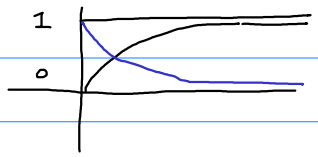


$$f(x) = \lambda e^{-\lambda x}$$

$$\mathbb{E}(x) = \int_0^\infty x \lambda e^{-\lambda x} dx$$

P. Distribution function $F_X(x) = P[X \le x]$

$$= 1 - e^{-\lambda x}$$



CCDF

CDF

(1) characteristic function of RV X

$$\phi_X(u) = \mathbb{E}\left[e^{jux}\right]$$

$$= \int_{-\infty}^{\infty} e^{jux} f_X(x)\, dx$$

$$f_X(x) = \lambda e^{-\lambda x}$$

$$\phi_X(u) = \int_{-\infty}^{\infty} e^{jux} \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^\infty e^{(ju-\lambda)x} dx$$

$$= \lambda \left[\frac{e^{(ju-\lambda)x}}{ju-\lambda}\right]_0^\infty$$

$$= \lambda \frac{[1-0]}{ju-\lambda} = \frac{-\lambda}{ju-\lambda}$$

$$= \lambda/(\lambda - ju)$$

(2) $$\left.\frac{d^n}{du^n}\phi_X(u)\right|_{@\ u=0} = j^n \mathbb{E}[x^n]$$

$$\left(\frac{\lambda}{\lambda - ju}\right) = \phi_X(u)$$

$$\boxed{\phi_X^{(n)}(u) = j^n \mathbb{E}[x^n]}$$

$$\phi_X(0) = \mathbb{E}[x^0] = 1$$

$$\phi_X^{(1)}(0) = j^1 \mathbb{E}[x] = j\bar{x} =$$

$$\phi_X^{(1)}(u) = \lambda \frac{d}{du}(\lambda - ju)^{-1}$$

$$= \lambda(-1)(\lambda - ju)^{-2}(-1)\, j$$

$$= \frac{j\lambda}{(\lambda - ju)^2}$$

$$\bar{x} = \frac{1}{\lambda}$$

$$\frac{j\lambda}{(\lambda - 0)^2} = \frac{j}{\lambda} = j\bar{x}$$

$$\therefore$$

$$\phi_X^{(2)}(u) = \frac{d}{du}\left[\frac{j\lambda}{(\lambda - ju)^2}\right] = j\lambda(-2)[\lambda - ju]^{-3}(-j) = \frac{-2\lambda}{(\lambda - ju)^3}$$

$$\therefore \quad j^2 \overline{x^2} = \frac{-2\lambda}{(\lambda)^3} \Rightarrow \overline{x^2} = \frac{2}{\lambda^2} = \mathbb{E}[x^2]$$

$$\text{Var}(X) = \mathbb{E}[x^2] - [\mathbb{E}[x]]^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Moment generating function $M_X(v) = \mathbb{E}[e^{vX}]$

$$= \int_0^\infty e^{vx} \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^\infty e^{-(\lambda - v)x} dx$$

$$= \frac{\lambda}{(\lambda - v)} \left[e^{-(\lambda - v)x}\right]_\infty^0$$

$$= \frac{\lambda}{\lambda - v} [1 - 0], \quad \lambda > v$$

$$= \frac{\lambda}{\lambda - v}, \quad \lambda > v$$

$$M_X^{(n)}(v) = \frac{d^n}{dv^n} M_X(v)$$

$$\boxed{M_X^{(n)}(0) = \overline{x^n} = \mathbb{E}[x^n]}$$

} Same results are obtained.

Let PDF of X be $F_X(x)$. Let pdf be denoted by $f_X(x)$

Consider the Laplace transform of $f_X(x)$?

$$L^*(s) = \mathbb{E}[e^{-sx}]$$ ← here s is complex

$$= \int_{-\infty}^{\infty} e^{-sx} e^{-\lambda x} \lambda \, dx$$

$$\boxed{L_X^{*(n)}(0) = (-1)^n \overline{x^n}}$$

Memoryless property. $\underbrace{P[X > s+t \mid X > t]}_{} = \underbrace{P[X > s]}_{F_X(s)}$

$$\frac{P[X > (s+t)]}{\bar{F}_X(t)} = \frac{P[X > (s+t),\ X > t]}{P[X > t]} = \bar{F}_X(s)$$

$$\therefore \quad \bar{F}_X(s+t) = \bar{F}_X(t) \cdot \bar{F}_X(s)$$

for exponential RV $\quad \bar{F}_X(x) = e^{-\lambda x}$

$$\therefore \quad \bar{F}_X(s+t) = e^{-\lambda(s+t)} = \underbrace{e^{-\lambda s}}_{\bar{F}_X(s)} \cdot \underbrace{e^{-\lambda t}}_{\bar{F}_X(t)}$$

$$\lambda(t) = \frac{f_X(t)}{1 - F_X(t)} = \frac{\lambda e^{-\lambda t}}{1 - [1 - e^{-\lambda t}]} = \lambda$$

$$P[X < t + dt \mid X > t] = P[t < X < (t+dt) \mid X > t]$$

$$= \frac{P[X \in [t, t+dt],\ X > t]}{P[X > t]}$$

$$= \frac{P[X \in [t, t+dt]]}{P[X > t]} \approx \frac{f_X(t)\ dt}{\bar{F}_X(t)}$$

$$= \underbrace{\frac{f_X(t)}{1 - F_X(t)} \cdot dt}$$

intensity of the conditional probability of lifetime

$$\lambda(t) = \frac{f_X(t)}{1 - F_X(t)} = \frac{\frac{d}{dt} F_X(t)}{1 - F_X(t)}$$

$$\int_0^t \lambda(t)\,dt = \int_{F_X(0)}^{F_X(t)} \frac{1}{1-F_X(t)}\,dF_X(b)$$

$$\int_0^t \lambda(x)\,dx = \int_{F_X(0)}^{F_X(b)} \frac{1}{1-F_X(x)}\,dF_X(x) = \int_0^{F_X(t)} \frac{1}{1-y}\,dy$$

$$= -\left[\ln(1-y)\right]_0^{F_X(t)}$$

$$dF_X(x) = dy$$

$$F_X(0) = y_0 = 0, \quad F_X(t) = y_t$$

$$-\int_0^t \lambda(x)\,dx = \ln\left[1-F_X(t)\right]$$

$$e^{-\int_0^t \lambda(x)\,dx} = 1-F_X(t)$$

$$F_X(t) = 1 - e^{-\int_0^t \lambda(x)\,dx}$$

Gamma distribution with parameters $\alpha, \beta > 0$:

$$f_X(x) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}$$

Suppose α is an integer [there]

$$f_X(x) = \frac{\beta^n x^{n-1} e^{-\beta x}}{\Gamma(n)} = \frac{\beta^n x^{n-1} e^{-\beta x}}{(n-1)!}$$

Characteristic function $\phi_X(u) = \left[1 - \frac{ju}{\beta}\right]^{-\alpha}$

$$= \left[1 - \frac{ju}{\beta}\right]^{-n}$$

$$= \left[\frac{\beta - ju}{\beta}\right]^{-n}$$

$\frac{\beta^n}{(\beta - ju)^n}$

Let $Z = \sum_{i=1}^n X_i$ (exponential)

$$\phi_Z(u) = \mathbb{E}\left[e^{juZ}\right]$$

$$= \mathbb{E}\left[e^{ju\sum_{i=1}^N X_i}\right]$$

$$= \mathbb{E}\left[e^{\sum_{i=1}^N juX_i}\right]$$

$$= \mathbb{E}\left[e^{juX_1} e^{juX_2} \cdots e^{juX_N}\right]$$

$$= \prod_{i=1}^N \mathbb{E}\left[e^{juX_i}\right]$$

$$= \prod_{i=1}^N \frac{\lambda}{\lambda - ju} = \frac{\lambda^N}{(\lambda - ju)^N}$$

Identical to the characteristic function of a Gamma distributed RV with parameters $N, \lambda > 0$

You may also use proof by induction Key ||

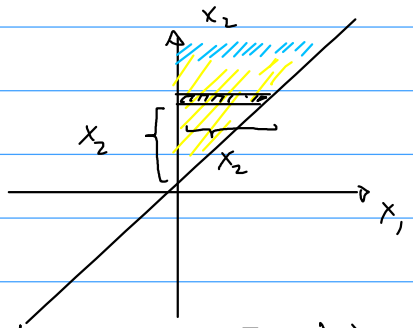
RV is are independent.

$Z = X + Y$

$$f_Z(z) = f_X * f_Y(z) = \int_{-\infty}^{\infty} f_X(z-u)\, f_Y(u)\, du$$

X_1 X_2 independent RVs exponential (λ_1 λ_2 parameters)

$P[X_1 \leq X_2]$:



$$P[X_1 \leq X_2] = \int_0^{\infty}\int_0^{X_2} \lambda_1\lambda_2 e^{-\lambda_1 x_1} e^{-\lambda_2 x_2}\, dx_1\, dx_2 = \lambda_1\lambda_2 \int_0^{\infty} e^{-\lambda_2 x_2} \int_0^{x_2} e^{-\lambda_1 x_1}\, dx_1\, dx_2$$

$$= \lambda_1\lambda_2 \int_0^{\infty} e^{-\lambda_2 x_2} \underbrace{\left[e^{-\lambda_1 x_1}\right]_0^{x_2}}_{\lambda_1} dx_2 = \frac{\lambda_1\lambda_2}{\lambda_1} \int_0^{\infty} e^{-\lambda_2 x_2} \left[1 - e^{-\lambda_1 x_2}\right] dx_2$$

$$= \lambda_2 \underbrace{\int_0^{\infty} e^{-\lambda_2 x_2}\, dx_2}_{=1} - \lambda_2 \int_0^{\infty} e^{-(\lambda_1+\lambda_2) x_2}\, dx_2$$

$$= 1 - \frac{\lambda_2}{(\lambda_1+\lambda_2)} \left[e^{-(\lambda_1+\lambda_2) x_2}\right]_0^{\infty} = 1 - \frac{\lambda_2}{(\lambda_1+\lambda_2)} [1 - 0]$$

$$= \frac{\lambda_1+\lambda_2-\lambda_2}{\lambda_1+\lambda_2} = \frac{\lambda_1}{\lambda_1+\lambda_2}$$ //

* Another way of computing $P[X_1 < X_2]$ is by conditioning on one variable. In this respect the mixed form of Bayes rule is very useful [see Wozencraft page 75]

RV: X — a joint density: $f_{XA}(a, A)$ — an event: A

$$P[A \mid X = a] = \frac{f_{XA}(a, A)}{f_X(a)}$$

$f_X(a)$: probability density of RV X

This is a joint density

Thus, we have $P[A \mid X = x]\, f_X(x) = f_{XA}(x, A)$ — (1)

By the def^n of joint densities we have

$$P[A] = \int_{-\infty}^{\infty} f_{XA}(x, A)\, dx \quad \text{— (2)}$$

From (1) & (2):

$$P[A] = \int_{-\infty}^{\infty} P[A \mid X = x]\, f_X(x)\, dx$$

In our previous example the event $A = X_1 \leq X_2$

Another method?

$$P[X_1 \leq X_2] = \int_0^{\infty} P[\underbrace{X_1 \leq X_2}_{A} \mid X_2 = x_2]\, \lambda_2 e^{-\lambda_2 x_2}\, dx_2$$

$\because$ independence

$$= \int_0^{\infty} P[X_1 \leq x_2]\, \lambda_2 e^{-\lambda_2 x_2}\, dx_2$$

$$= \int_0^{\infty} F_{X_1}(x_2)\, \lambda_2 e^{-\lambda_2 x_2}\, dx_2$$

$$= \lambda_2 \int_0^{\infty} (1 - e^{-\lambda_1 x_2})\, e^{-\lambda_2 x_2}\, dx_2$$

$=$ proceed to get $\frac{\lambda_1}{\lambda_1 + \lambda_2}$

$$P[X_1 = \min(X_1, X_2)] = P[X_1 \leq X_2] = \frac{\lambda_1}{\lambda_1 + \lambda_2} \quad //$$

RV

$$Z = \min_i \{X_i\} \longrightarrow P[Z \leq z] \qquad \text{we are done !}$$

$$P[Z \leq z] = 1 - P[Z > z]$$

$$= 1 - P[\min_i X_i > z]$$

$$= 1 - P[\text{all } X_i > z]$$

$$= 1 - P[X_1 > z, X_2 > z, \ldots, X_N > z]$$

$$= 1 - \prod_{i=1}^{N} P[X_i > z]$$

$$P[Z \leq z] = 1 - \prod_{i=1}^{N} e^{-\lambda_i z} = 1 - e^{-\left(\sum_{i=1}^{N} \lambda_i\right) z}$$

CDF of a $\begin{cases} \text{Exp} \\ \text{RV} \end{cases}$ with parameter $\lambda = \sum_{i=1}^{N} \lambda_i$

$$Z = \max_i \{X_i\}$$

$$P[Z \leq z] = P[\max_i \{X_i\} \leq z]$$

$$= P[\text{All } X_i \leq z] = P[X_1 \leq z, X_2 \leq z, \ldots, X_N \leq z]$$

$$F_Z(z) = \prod_{i=1}^{N} P[X_i \le z] = \prod_{i=1}^{N} \left(1 - e^{-\lambda_i z}\right)$$

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \text{complicated}$$

Identical RVs $\Rightarrow$ $F_Z(z) = \left(1 - e^{-\lambda z}\right)^N$

$$f_Z(z) = N \left(1 - e^{-\lambda z}\right)^{N-1} \cdot (-) e^{-\lambda z} \cdot (-\lambda)$$

$$= \lambda N \frac{\left(1 - e^{-\lambda z}\right)^{N-1}}{e^{-\lambda z}} \quad ?$$

$P[X_i = \min\{X_1, X_2, \dots X_n\}]$

$= P\left[X_i \le \min\{X_1 X_2 \cdots X_{i-1} X_{i+1} X_{i+2} \cdots X_n\}\right]$

exponentially distributed with

$\lambda = (\lambda_1 + \lambda_2 + \cdots + \lambda_{i-1} + \lambda_{i+1} + \cdots \lambda_n)$

$$= \frac{\lambda_i}{\lambda_i + \lambda} = \frac{\lambda_i}{\sum_{j=1}^{n} \lambda_j}$$

If identical $\Rightarrow$ The above probability

$$= \frac{\lambda}{n\lambda} = \frac{1}{n}$$

failure rate: $\lambda(t) = \dfrac{f(t)}{1 - F(t)}$

Interpret via $P\left[X \in [t, t+dt] \mid X > t\right]$

$$= P\left[t < X < t+dt \mid X > t\right]$$

$$= \frac{P\left[X \in [t, t+dt], X > t\right]}{P[X > t]}$$

$$= \frac{P\left[X \in [t, t+dt]\right]}{1 - F(t)} \approx \frac{f_X(t)}{1 - F(t)} \times dt$$

this is the intensity of conditional pdf of X given $X > t$, at $x = t$

see next page

* If one needs to compute $P[X \in A \mid X \geq t]$

it is computed by using the conditional pdf of X given $X > t$.

↳ i.e., $f_X(x \mid X>t)$

$$P[\underbrace{X \in [t, t+dt]}_{\text{our event } A} \mid X>t] = \int_t^{t+dt} f_X(x \mid X>t)\, dx$$

$$\approx \underbrace{f_X(t \mid X>t)}\, dt$$

$$= \frac{f_X(t)}{1-F_X(t)} = \lambda(t)$$

See Wozencraft page 75 { here equality is from the mixed form of Bayes rule

$$\underbrace{f_X(x \mid X>t)}_{\text{same as } f_X(x \mid X \geq t)} = \frac{f_X(x, A)}{P[X \geq t]} \quad \text{event } \{X>t\}$$

[no probability mass at $X=t$]

$$= \frac{P[A \mid X=x]\, f_X(x)}{P[X \geqslant t]} = \frac{P[X \geqslant t \mid X=x]\, f_X(x)}{1-F(t)}$$

* How to compute ?

[check Wozencraft eq: (2.1026)]

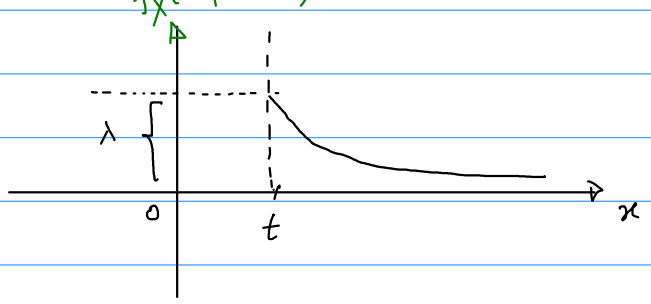
$$P[X \geqslant t \mid X=x] = \begin{cases} 1 & x \geqslant t \\ 0 & x < t \end{cases}$$

Thus $\underbrace{f_X(x \mid X \geqslant t)}_{= f_X(x \mid X>t)} = \begin{cases} f_X(x) \,/\, 1-F(t) & x \geqslant t \\ 0 & \text{otherwise} \end{cases}$

e.g., for exponential distribution

$$f_X(x \mid X \geqslant t) = \frac{f_X(x)}{e^{-\lambda t}} = \lambda \frac{e^{-\lambda x}}{e^{-\lambda t}}, \quad x \geqslant t$$

$$= \underbrace{\lambda \cdot e^{-\lambda(x-t)}} \quad x \geqslant t$$



Simply the shifted version of the original exponential

RESTART AGAIN given that $X \geqslant t$ ≡ Memoryless property.